

Methodology – Michigan Mobility Supply Chain Staffing Patterns (v0.1)

Project: Michigan Automotive & Mobility Supply Chain – Occupational Staffing Patterns & Forecasts

Repo root: EV-Transition/

Primary outputs: processed staffing tables (MI & US), segment/stage rollups, 2024→2030 tables & figures, Streamlit dashboards

1) Objectives & Scope

- Build a reproducible pipeline to quantify **occupational staffing patterns by industry** for Michigan's automotive supply chain and compare to U.S. benchmarks.
 - Organize industries into **10 supply-chain segments** grouped by **Stage** (Upstream / OEM / Downstream). Add **segment 0** = "All Segments" for statewide totals.
 - Produce **baseline (2024)** and **projection** views (to 2030 and 2034), exploring **four methodology combinations**:
 - Attribution of core-auto totals: **Lightcast** vs **BEA** (shares applied to QCEW)
 - Growth of segment totals: **BLS** vs **Moody's** (segment-level employment trajectories)
 - Preserve **Michigan's 2024 composition** while incorporating **national directional changes** in staffing patterns via a delta method.
 - Deliver artifacts for collaborators:
 - interim & processed datasets,
 - executed notebooks with figures,
 - dashboards (v1 operational; v2 walkthrough).
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2) Repository Layout (relevant parts)

```
EV-Transition/
├── config/
│   ├── project.yml
│   ├── segments.yml
│   └── colors.json
├── data/
│   ├── raw/
│   │   ├── NAICS_sectors_August_18_2025.xlsx
│   │   ├── QCEW_MI_2024.csv
│   │   ├── occupation_2024_ep/ (BLS EP tables incl. Table 1.2, 1.9)
│   │   └── us_staffing_patterns/ (38 CSVs, one per NAICS; fallbacks noted)
│   ├── lookups/
│   │   └── segment_assignments.csv (NAICS→segment_id, segment_name, stage, MI
```

QCEW 2024)

- | | | interim/
 - | | | | moodys_us_2024_2030.csv
 - | | | | moodys_michigan_2024_2030.csv
 - | | | | moodys_us_stages_2024_2030.csv
 - | | | | moodys_michigan_stages_2024_2030.csv
 - | | | | moodys_us_segments_2024_2030.csv
 - | | | | moodys_michigan_segments_2024_2030.csv
 - | | | | occupation_table12_clean.xlsx
 - | | | | occupation_table12_tidy.csv
- | | | processed/
 - | | | | us_staffing_segments_summary.csv
 - | | | | us_staffing_segment_sources.csv
 - | | | | us_staffing_segments_flags.csv
 - | | | | occupation_table12_comparison.xlsx / .csv
 - | | | | mcda_staffing_major_2021_2024.csv
 - | | | | mcda_staffing_detailed_2021_2024.csv
 - | | | | mcda_staffing_long_2021_2024.xlsx
 - | | | | mcda_staffing_education_summary.csv
 - | | | | mi_occ_segment_totals_2024_2030.csv
 - | | | | mi_occ_segment_totals_2024_2034.csv
 - | | | | mi_occ_segment_totals_validation.csv
 - | | | | mi_qcew_segment_employment_timeseries_coreauto_extended_compare.csv
- | | notebooks/
 - | | | 03_growth_trajectories.ipynb (+ _executed)
 - | | | 04_mcda_staffing_analysis.ipynb (+ _executed)
 - | | | 14_occupation_forecasts.ipynb
 - | | | 15_occupation_change_tables.ipynb
- | | scripts/
 - | | | update_segment_lookup.py
 - | | | process_moodys.py
 - | | | aggregate_moodys_segments.py
 - | | | process_occupation_table12.py
 - | | | process_mcda_staffing.py
 - | | | fetch_us_staffing.py
 - | | | process_us_staffing_segments.py
 - | | | occupation_forecasts_from_segment_totals.py
- | | dashboards/
 - | | | occupation_forecast_dashboard.py (v1)
 - | | | occupation_forecast_dashboard_v2.py (annotated walkthrough)
- | | reports/
 - | | | figures/
 - | | | summaries/
 - | | | analysis_progress_report.md

3) Data Sources

3.1 Supply-chain structure

- `data/raw/NAICS_sectors_August_18_2025.xlsx`: 38 four-digit NAICS mapped to Stage $\in$ {Upstream, OEM, Downstream} and to **10 segments**. Segment 1 has materials sub-groups; canonical labeling standardizes to a single segment name per `segment_id`.
- **Lookup**: `data/lookups/segment_assignments.csv` holds cleaned titles, `segment_id`, `segment_name`, and `stage` + **MI QCEW 2024 employment** appended from `QCEW_MI_2024.csv`.

3.2 Michigan employment & staffing

- **QCEW MI 2024**: `data/raw/QCEW_MI_2024.csv` – employment counts by NAICS (used to seed segment totals; later adjusted by attribution shares).
- **MCDA staffing (MI)**: historical staffing patterns by segment (major & detailed occupations) for **2021 & 2024**.
- Processed by `scripts/process_mcda_staffing.py` → `data/processed/mcda_*`. Includes EP (BLS Table 1.2) joins and **education groups** for detailed SOCs (HS or less / Some college or Associate / BA+).

3.3 U.S. staffing benchmarks

- **BLS “Table 1.9” links**: industry→occupation matrices for national staffing patterns.
- Scraped via `scripts/fetch_us_staffing.py` into `data/raw/us_staffing_patterns/`.
- **Fallbacks** where BLS lacks 4-digit landings (e.g., 3272→327000, 4841/4842→484000, etc.) are recorded. Deduplication prevents double-counting when multiple NAICS share the same broader matrix (e.g., Truck Transportation).
- **Segment rollups (US)**: `scripts/process_us_staffing_segments.py` aggregates NAICS tables to the 10 segments → `data/processed/us_staffing_segments_summary.csv` + sources/flags.

3.4 Growth paths for segment totals

- **Moody’s**: workbook ingested with `scripts/process_moody’s.py` to compute **2024→2030 %** changes for Employment / Wages / GDP (US & MI). Handle **4471→4571** mapping for Gasoline Stations in MI.
- **Segment & Stage aggregates**: `scripts/aggregate_moody’s_segments.py` → `data/interim/moody’s*_segments_2024_2030.csv` and `moody’s*_stages_2024_2030.csv`.
- **BLS**: not a direct time series here; instead, BLS **national staffing deltas** are used to guide **mix shifts** (see §5). BLS EP **Table 1.2** 2023/2024 used for projections metadata & comparisons.

4) Baseline Attribution of Core Auto (segment totals)

Segment totals are initialized from MI QCEW and then attributed to **Core Auto** via external shares:

- **Lightcast attribution**

- Raw: `data/raw/auto_attribution_core_auto_lightcast.csv` (column `share_to_set`).
- Applied in `scripts/apply_lightcast_share_and_extend.py` → produces adjusted core-auto series.

• BEA attribution

- Raw: `data/raw/auto_attribution_bea.csv` (column `bea_share_to_set`).
- Applied in `scripts/apply_bea_share_and_extend.py` → produces adjusted core-auto series.

Both flows multiply QCEW segment employment by the respective `share_to_set` to set an **attributed baseline**. These baselines feed the **four methodology combinations** when combined with growth paths below.

5) Staffing Pattern Evolution (mix-shift methodology)

Goal: Keep **Michigan's 2024 composition** but allow **directional changes** from U.S. patterns. We **do not force** MI to equal US by 2034.

5.1 Notation

- Segment s , occupation i , year $t \in [2024, 2034]$.
- **Michigan base share (2024):** $b_{i,s} = \text{MCDA share}_{i,s,2024}$.
- **BLS national shares:** $u_{i,s,2024}, u_{i,s,2034}$ from `us_staffing_segments_summary.csv`.

5.2 Growth factor and targets

- **BLS growth factor** (guarded):

$$g_{i,s} = \begin{cases} \frac{u_{i,s,2034}}{u_{i,s,2024}} & \text{if } u_{i,s,2024} > 0 \\ 1 & \text{otherwise (fallback)} \end{cases}$$
- **Raw target share (un-normalized):** $\tilde{t}_{i,s} = b_{i,s} \cdot g_{i,s}$.
- **Normalize within segment:** $t_{i,s} = \tilde{t}_{i,s} / \sum_j \tilde{t}_{j,s}$.

5.3 Interpolation across years

- Linear weight: $w_t = \frac{t-2024}{2034-2024} \in [0, 1]$.
- **Interpolated share (pre-normalization):** $\tilde{p}_{i,s,t} = b_{i,s} [1 + (g_{i,s} - 1) w_t]$.
- **Re-normalize each year:** $p_{i,s,t} = \tilde{p}_{i,s,t} / \sum_j \tilde{p}_{j,s,t}$.

Edge cases: if any $u_{i,s,2024} = 0$ or missing, set $g_{i,s} = 1$ (no pressure from US) so Michigan holds its base composition locally but the segment still renormalizes.

5.4 Segment totals and segment 0

- Segment totals by year $E_{s,t}$ come from the chosen **attribution** (Lightcast/BEA) × **growth path** (Moody's/BLS).

- **Occupation employment:** $\hat{E}_{i,s,t} = p_{i,s,t} \cdot E_{s,t}$.
- After all segments are computed, a synthetic **segment 0 (All Segments)** is created by summing $\hat{E}_{i,s,t}$ over $s = 1..10$, then **recomputing shares** so totals exhaust statewide employment.

Implementation reference: `scripts/occupation_forecasts_from_segment_totals.py` (functions `interpolate_shares`, segment 0 post-processing).

6) U.S. Segment Rollups & Fallback Rules

Produced by `scripts/process_us_staffing_segments.py`:

- Iterate over **unique** `segment_id` to avoid duplicate sublabels for segment 1.
 - **Deduplicate shared BLS sources** within a segment: if 4841 and 4842 both point to 484000, use the 484000 matrix once.
 - **Flags & sources:**
 - `us_staffing_segment_sources.csv` shows NAICS→source mapping (e.g., 3272→327000).
 - `us_staffing_segments_flags.csv` highlights re-used sources across segments (e.g., 3320A1 spanning two segments) for analytical caution.
 - Segment share computation uses **00-0000 (Total, all occupations)** where available as denominators (otherwise sums detailed rows). This is documented and consistent across segments.
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7) Comparative MI vs. US Staffing Patterns

- `scripts/compare_us_mi_segments.py` aligns MI and US by `(segment_id, occcd)` and restricts **major** to `-0000` codes to match MI “Summary” groups; excess U.S. summaries are dropped.
- Outputs:
- `us_mi_segment_comparison_major.csv`
- `us_mi_segment_comparison_detailed.csv`
- `us_mi_segment_comparison_flags.csv` (legitimate asymmetries; MI suppressions; U.S.-only occupations)

Interpretation: Differences are expected to be **minor** for many segments; major mismatches typically trace to **fallback matrices** or **Michigan-specific composition**.

8) Notebooks & Visuals

- **03_growth_trajectories.ipynb:** MI vs US **stage & segment** % change bars (Employment/Wages/GDP) for 2024→2030; configured palette (`colors.json`) with Michigan **teal** and U.S. **blue**.
- **04_mcda_staffing_analysis.ipynb:** 2021→2024 MI staffing shifts (major & detailed), education mix changes.
- **14_occupation_forecasts.ipynb:** reads preferred `mi_occ_segment_totals_*.csv` forecasts; introduces `has_bls_shift` flag; uses **2030 snapshot** for reporting.

- **15_occupation_change_tables.ipynb**: tables for 2024→2030 **employment deltas** by segment and by education across segments; highlights **methodology spreads**.

All generated figures saved under `reports/figures/`. A slide deck is programmatically built by `scripts/create_figures_presentation.py` (optional theming from `related-scripts-output/*.pptx`).

9) Dashboard (v1) – Key Features

- **Overview**: 2024→2030 cards per selected methodology; duplicate set for **Upstream & Core Auto (1–7)** excluding 8–10. Primary UI color overridden to teal where supported.
- **Segments**: bar charts + tables by segment (excludes segment 0 in charts).
- **Stage/Horizon**: forecast lines; **extended core context (2001–2055)** responsive to current segment selection.
- **Occupation Explorer**: filter by segment & education; sort by Employment 2024, 2030, Δ , or $\% \Delta$.
- **Supply Chain**: live table from `segment_assignments.csv` with QCEW 2024 employment.

Launch locally: `streamlit run dashboards/occupation_forecast_dashboard.py`

Optional auth: add `streamlit-authenticator` with hashed credentials.

10) Validation & QA

1. **Consistency checks**
2. Within each `(segment_id, year, methodology)`, confirm **shares sum to 1.0** ($\pm 1e-6$).
3. Segment 0 totals equal the sum of segments 1–10 for each `(year, methodology)`.
4. **BLS delta sanity**
5. When $u_{i,s,2024} = 0$, verify growth factor = 1.0 and MI base share is preserved up to normalization.
6. **US vs MI comparison**
7. Spot-check a few segments (e.g., **7. Core Auto**) for large share gaps; investigate **fallback matrices** first.
8. **Moody's harmonization**
9. **4471→4571** mapping applied before joining to MI lookup; audit the mirrored values in `moodys_michigan_2024_2030.csv`.
10. **EP projection joins**
11. Major occupations now **retain EP fields** in `mcda_staffing_major_2021_2024.csv` (logic relaxed from earlier mask).

11) Known Limitations & Assumptions

- **Fallback matrices (US)**: Use of broader NAICS groups (e.g., 327000, 484000) may introduce bias; flags identify where caution is warranted.
- **Michigan suppressions**: Some detailed SOC's may be absent in MI but present in US; flagged in comparison outputs.

- **Linear interpolation** of mix shifts is a simplification; alternative dynamics (e.g., logistic adjustment) could be explored.
- **Attribution shares** (Lightcast/BEA) are treated as inputs; errors or vintage changes feed directly to attributed totals.

12) Reproducibility & Environment

- **Python** ≥ 3.10 ; core libs pinned via `requirements.txt` (e.g., `streamlit`, `pandas`, `numpy`, `plotly`).
- Optional: `streamlit-authenticator`, `python-pptx` for slides.
- Execute scripts from repo root; all paths are **relative**.
- Notebook execution via `nbconvert` for CI-style runs.

Example rebuild order (typical):

```
python scripts/update_segment_lookup.py
python scripts/process_moodys.py
python scripts/aggregate_moodys_segments.py
python scripts/process_occupation_table12.py
python scripts/process_mda_staffing.py
python scripts/fetch_us_staffing.py
python scripts/process_us_staffing_segments.py
python scripts/occupation_forecasts_from_segment_totals.py
python -m nbconvert --to notebook --execute notebooks/
03_growth_trajectories.ipynb
python -m nbconvert --to notebook --execute notebooks/
04_mda_staffing_analysis.ipynb
python -m nbconvert --to notebook --execute notebooks/
14_occupation_forecasts.ipynb
python -m nbconvert --to notebook --execute notebooks/
15_occupation_change_tables.ipynb
```

13) Data Dictionaries (selected)

13.1 `data/lookups/segment_assignments.csv`

- `naics` (str, 4-digit)
- `naics_title` (str)
- `segment_id` (int: 1–10; 0 reserved for totals)
- `segment_name` (str canonical)
- `stage` (Upstream/OEM/Downstream)
- `employment_qcew_2024` (int)

13.2 data/processed/us_staffing_segments_summary.csv

- Keys: `segment_id`, `Occupation Code`, `Occupation Title`, `Occupation Type` (Summary/Line item)
- Metrics: `2024 Employment`, `Projected 2034 Employment`, `segment_share_2024`, `segment_share_2034`
- Denominator preference: `00-0000` when available.

13.3 data/processed/mcda_staffing_*_2021_2024.csv

- Keys: `segment_id`, `occcd` (SOC), `occ_title`, `occ_level` (major/detailed)
- Time: `empl_2021`, `empl_2024`, deltas; shares within segment (major & detailed)
- EP fields joined for major & detailed: `ep_employment_2024`, `ep_change_percent`, `ep_openings_*`, etc.
- Detailed only: `ep_edu_grouped` (HS or less / SC or associate's / BA+).

13.4 data/processed/mi_occ_segment_totals_2024_2030.csv

- Keys: `segment_id` $\in \{0..10\}$, `occcd`, `soctitle`, `methodology` $\in \{\text{bea_bls, bea_moody, lightcast_bls, lightcast_moody}\}$, `year`
- Metrics: `employment`, `share`, `base_share_2024`, `target_share_2034` (+ validation sums by segment-year).
- Segment 0 is post-hoc rollup over segments 1–10.

14) Communication Aids

- **One-pagers:**
- MI vs US segment share comparison (major & detailed) with callouts where $|\Delta| > \text{threshold}$.
- 2024→2030 segment employment deltas under default methodology (lightcast_moody) and spread vs. alternatives.
- **Slides:** Auto-generated deck of all `reports/figures/` with themed template.
- **Dashboard v2:** Same analytics, with on-tab prose for stakeholders.

15) Next Steps

1. **Finalize segment definitions** text in `docs/segment_definitions.md` (plain-English descriptions; rationale for sub-groups in Segment 1).
2. **Document fallbacks** explicitly in `docs/methodology.md` §US Staffing Benchmarks, linking to `us_staffing_segment_sources.csv` & flags.
3. **Decide handling** for any MI–US gaps (e.g., suppressions) in comparative reporting (drop, impute, or annotate).
4. Extend forecast **to 2034** visuals alongside 2030 snapshot; add export hooks for client-ready tables.
5. Optional: test **nonlinear** share transitions as sensitivity (e.g., partial adoption of US deltas by segment).

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Last updated: *auto-generated in draft; update when committing*